trishna_ORDERING

TRISHNA urinal hydration monitor: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: reagentless SG estimate needs per-user refractometer calibration; wellness trend only, not a medical device. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/trishna/1_fabrication/trishna_ctrl_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/trishna/2_assembly/trishna_ctrl_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/trishna/2_assembly/trishna_ctrl_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/trishna_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J1	12 V DC in (2 A)	CUI/Same Sky PJ-063AH	C22434582
TP1, TP2, TP3, TP4	TP		
J4	DevKitC left: 3V3 EN 36 39 34 35 32 33 25 26 27 14 12 GND 13 D2 D3 CMD 5V	BOOMELE 2.54-1*19P female header	C319202
J5	DevKitC right: GND 23 22 TX RX 21 GND 19 18 5 17 16 4 0 2 15 D1 D0 CLK	BOOMELE 2.54-1*19P female header	C319202
JP1	3V3 link (remove for USB-only power)	XFCN PZ254V-11-02P + jumper	C492401
J6	Peristaltic pump 12 V (red = PUMP_O1)	JST B2B-PH-K-S(LF)(SN)	C131337
J7	Solenoid valve 12 V NC	JST B2B-PH-K-S(LF)(SN)	C131337
J8	Sensor-block LEDs (450 nm blue, 850 nm IR)	JST B4B-PH-K-S(LF)(SN)	C131334
J9	Photodiodes BPW34: transmission, 90-degree scatter (anodes to GND)	JST B3B-PH-K-S(LF)(SN)	C131339
J10	EC electrodes (titanium pins E1, E2)	JST B2B-PH-K-S(LF)(SN)	C131337
J11	Sump stream-detect probe	JST B2B-PH-K-S(LF)(SN)	C131337
J12	Chamber-B floor leak probe	JST B2B-PH-K-S(LF)(SN)	C131337
J13	DS18B20 temperature probe	JST B3B-PH-K-S(LF)(SN)	C131339
J14	VL53L1X ToF presence sensor (I2C)	JST B4B-PH-K-S(LF)(SN)	C131334
J15	Telemetry UART (ESP32 TX out, RX in) to USB-serial	JST B3B-PH-K-S(LF)(SN)	C131339

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US9.16*for1board* * *, * * *US*8.23 each for 10 boards**, **US$6.45 each for 100 boards** (about INR 806 / 724 / 568 at 88 INR/USD). The per-line break-up is in `costing/trishna_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
J1	1	12 V DC in (2 A)	CUI/Same Sky PJ-063AH	C22434582	2.7602	2.760
F1	1	PTC 2A 16V	R+O SMD1812-200-16 (2 A hold)	C46641167	0.0771	0.077
D3	1	SMBJ15A	Littelfuse SMBJ15A	C83846	0.1300	0.130
Q5	1	AO3401A	AO3401A	C15127	0.0969	0.097
D4	1	BZT52C12	DIODES BZT52C12-7-F	C124196	0.0775	0.077
R40,R61,R71,R86,R87,R88,R89	7	10k	0603 10k 1%	C25804	0.0024	0.017
C30,C40	2	100u/35V	Lelon VZH101M1VTR-0607	C176683	0.0881	0.176
C31,C41	2	10u/50V	Samsung CL32A106KBJNNNE	C380537	0.2086	0.417

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U2	1	AP63203WU-7	AP63203WU-7 (3.3 V fixed)	C780769	1.2058	1.206
R41,R60,R64,R66,R69	5	100k	0603 100k 1%	C25803	0.0030	0.015
C32,C37,C60,C39,C42,C65,C70	7	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.052
L1	1	4.7u	Changjiang FNR5040S4R7MT	C167963	0.0933	0.093
C33,C34	2	22u/10V	Samsung CL21A226MPQNNNE	C29277	0.0377	0.075
FB1	1	BLM18 600R	Murata BLM18AG601SN1D	C19330	0.0122	0.012
C36,C38,C69	3	10u	0805 10uF X5R	C15850	0.0441	0.132
D6	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011
R45,R73,R74,R79,R81	5	1k	0603 1k 1%	C21190	0.0017	0.008
TP1,TP2,TP3,TP4	4	TP				
J4	1	DevKitC left: 3V3 EN 36 39 34 35 32 33 25 26 27 14 12 GND 13 D2 D3 CMD 5V	BOOMELE 2.54-1*19P female header	C319202	0.1509	0.151
J5	1	DevKitC right: GND 23 22 TX RX 21 GND 19 18 5 17 16 4 0 2 15 D1 D0 CLK	BOOMELE 2.54-1*19P female header	C319202	0.1509	0.151
JP1	1	3V3 link (remove for USB-only power)	XFCN PZ254V-11-02P + jumper	C492401	0.0189	0.019
U3	1	DRV8871DDAR	TI DRV8871DDAR	C75864	1.6982	1.698
R62,R80	2	47k	UNI-ROYAL 0603WAF4702T5E	C25819	0.0020	0.004
J6	1	Peristaltic pump 12 V (red = PUMP_O1)	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
Q1,Q2,Q3	3	AO3400A	AO3400A	C20917	0.0899	0.270
R63,R65,R68,R76,R77	5	100	UNI-ROYAL 0603WAF1000T5E	C22775	0.0024	0.012
D10	1	M7 (1N4007)	MDD M7 (1N4007, 1 A 1000 V)	C95872	0.0104	0.010
J7	1	Solenoid valve 12 V NC	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
R67,R70	2	1k 1/4W	UNI-ROYAL 1206W4F1001T5E	C4410	0.0086	0.017
J8	1	Sensor-block LEDs (450 nm blue, 850 nm IR)	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
U4	1	MCP6002T-I/SN	Microchip MCP6002T-I/SN	C7377	0.3121	0.312
C61,C63,C64	3	10n	Samsung CL10B103KB8NNNC 0603 10nF X7R 50V	C1589	0.0075	0.022
R72	1	1M	UNI-ROYAL 0603WAF1004T5E	C22935	0.0018	0.002
C62,C66,C67,C68	4	1n	Samsung CL10B102KB8NNNC X7R 50V	C1588	0.0099	0.040

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
J9	1	Photodiodes BPW34: transmission, 90-degree scatter (anodes to GND)	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
R75	1	330	UNI-ROYAL 0603WAF3300T5E	C23138	0.0021	0.002
J10	1	EC electrodes (titanium pins E1, E2)	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
R78,R82,R83,R84	4	4.7k	0603 4.7k 1%	C23162	0.0028	0.011
J11	1	Sump stream-detect probe	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
J12	1	Chamber-B floor leak probe	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
J13	1	DS18B20 temperature probe	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J14	1	VL53L1X ToF presence sensor (I2C)	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
J15	1	Telemetry UART (ESP32 TX out, RX in) to USB-serial	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J16	1	microSD (SPI log)	Molex 104031-0811	C585350	0.6848	0.685

3. Bought-in and mechanical items

5.2 Bought-in items (India sources; prices are rough guides, check before ordering)

Item	Qty	Spec / example	Where	Approx. INR
ESP32-DevKitC-32E	1	**38-pin, 1.0 in (25.4 mm) row spacing**, WROOM-32E or 32D (not WROVER: GPIO16/17 are used). 30-pin DOIT boards do not fit	element14 India, Robu.in	900-1,100 (clones 450)
12 V DC adapter	1	12 V 2 A, 5.5 x 2.1 mm, centre positive, BIS-marked	local, Amazon.in	300-450
Peristaltic pump	1	12 V DC, about 0.2-0.4 A, 3 x 5 mm tube, 40-100 mL/min (e.g. Kamoer NKP-DC-S10B, or generic "12V DC dosing pump")	Robu.in, Amazon.in	450-2,500
Solenoid valve	1	12 V DC **normally closed, direct-acting, 0 MPa minimum**, <= 6 W, 1/8 in or 6-8 mm barbs, food-grade body (water-dispenser / coffee-machine type). **Not** an RO pilot valve	Amazon.in, Robu.in	250-600
VL53L1X ToF breakout	1	3.3 V, I2C (Pololu #3415 or CJMCU-531)	Robu.in, Robocraze	600-900
DS18B20 waterproof probe	1	6 x 50 mm stainless sheath, 1 m 3-wire cable	Robu.in	120-200
Photodiodes	2 (+1)	**OSRAM SFH 203** (5 mm clear, 400-1100 nm, +/- 75 deg); SFH 203 P also fits	element14 India, Mouser India	60-90 each
LED 450 nm	1 (+1)	5 mm water-clear blue, 30 deg or wider	local, Robu.in	10
LED 850 nm	1 (+1)	5 mm IR, 850 nm, 30 deg or wider (e.g. Vishay TSHG6200, or generic 850 nm)	element14, Robu.in	15-60

Item	Qty	Spec / example	Where	Approx. INR
Titanium wire	30 cm	Grade 2, 1.0 mm	Amazon.in (anodising / jewellery wire)	300-500
Silicone tubing	2 m + 2 m	food-grade 3 x 5 mm and 4 x 6 mm	Amazon.in, aquarium shops	150 each
M5 to 4 mm hose-barb fittings	6	**PP or nylon** (no plain brass in urine), M5 x 0.8 male	Amazon.in, pneumatics shops	40 each
Check valve	2	4 mm aquarium duckbill (one spare)	aquarium shops	30 each
Tee 4 mm	2	aquarium PP tee	aquarium shops	20 each
Reducers	2	6-8 mm to 4 mm barb (valve to tubing)	aquarium / Amazon.in	30 each
Cable glands	4 x M16 + 10 x M12	nylon, IP68, M16 x 1.5 (4-8 mm) / M12 x 1.5 (3-6.5 mm)	Robu.in, local electrical	kit 300
O-rings	4 x 5 x 1, 1 x 6 x 1.5	NBR or silicone	metric O-ring kit (Amazon.in)	250
Silicone cord	1.5 m	2.5 mm solid	Amazon.in	200
Heat-set inserts	40	M3 x 5.7 for 4.0 mm holes	Robu.in	250
Screws	-	22 x M3 x 8 (lids), 4 x M3 x 6 (PCB), 8 x M3 x 8 (brackets), 4 x M3 x 45 + nuts (block), 2 x M3 x 10 A2 stainless + 2 ring terminals (leak probe), 10 x M2 x 6 self-tapping (retainers), 8 wall screws + plugs	hardware shop	300
Cables	-	1 m 4-core 0.5 mm2; 1 m 8-core 0.22 mm2 alarm cable; 1 m 2-core shielded; 3 m 2-core 0.22 mm2; 2 m 4-core 0.22 mm2	local electrical	300
JST PH kit	1	PH 2.0 housings 2/3/4-pin + SPH-002T-P0.5S crimps	Robu.in	250
USB-serial adapter	1	CP2102 or CH340, 3.3 V logic	Robu.in	150
microSD	1	8-32 GB, FAT32	any	300
Reservoir	1	1.5-2 L food container + 4 mm aquarium bulkhead	local	200
Sealants	-	black RTV silicone (neutral cure), 2-part epoxy (Araldite Standard), PTFE tape	hardware shop	400
Calibration	-	KCl standards 1413 uS/cm and 12.88 mS/cm (Hanna HI7031 / HI7030 sachets) or self-made (section 9.2); handheld **urine-SG refractometer** with ATC (1.000-1.050)	lab suppliers, Amazon.in	600 + 900-1,500
Bench fluids	-	NaCl, KCl, urea (lab grade or 46 % N fertiliser urea for bench tests), riboflavin tablet or yellow food colour, citric acid	chemist, grocery	200
Filament	1 kg	black PETG	Robu.in, local	1,200

4. Confidence (read before you order)

- **Order confidence: Medium-High**: limited stock on: J1 12 V DC in (2 A) (stock 321); U2 AP63203WU-7 (stock 195).
- **Works-as-intended confidence: Low-Medium**: Controller is a standard ESP32 carrier (buck, H-bridge, MOSFET + flyback, TIAs); firmware compiles for the ESP32 and 19 host tests cover the frame, filters, EC maths and every state-machine path; CAD solids give a 2.02 mL chamber. Unproven until built: the printed

flow cell must be leak-free, the bipolar GPIO EC drive and the photodiode gains need tuning on the bench, and the SG estimate from conductivity + colour is only as good as the per-user refractometer fit.

- **First thing to do:** Print and pressure-test the sensor block, then run validation V1-V4 with the synthetic-urine fluids before any real use.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.